



CRYSTAL GRAPHICS DOCUMENT

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1 Physics

1.1 Light Transport

This section talks about how **physics** explain the behavior of light transport on a **macro** level. Sometimes we will dive to micro levels to explain how certain patterns develop but really we only care about what we can perceive as behaviors on a macro level.

1.1.1 Surface Level

This section talks about what happens when light meets the surface of objects. Notice the object surface we talk about here can be either surface of solids, or the surface of a soap bubble. So it can be more precisely described as positions with sudden material changes.

In totality, these macro properties effect what happens when light hits the surface:

- Index of Refraction (a.k.a. IOR)
- Microgeometry (Microgeometry doesn't technically effect how light is reflected / refracted on a micro level but its characteristics determine the macro behavior of a surface.)

The micro behavior of surface light transport can be described with Snell's Law & Fresnel Equations:

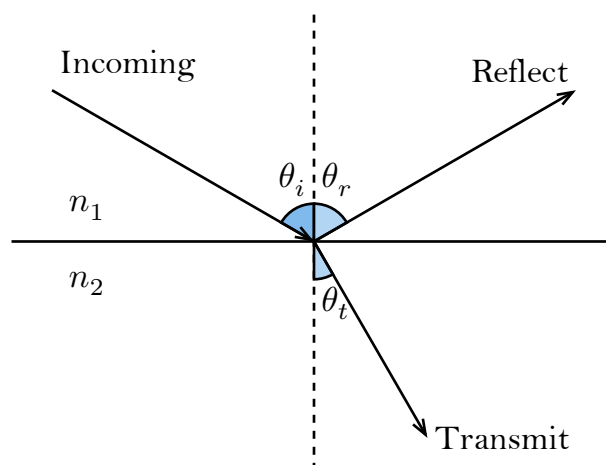


Figure 1: Micro Surface Level Light Transport

Snell's Law

$$\begin{cases} \frac{n_1}{n_2} = \frac{\sin(\theta_i)}{\sin(\theta_t)} \\ \theta_i = \theta_r \end{cases}$$

Fresnel Equations

$$\begin{cases} R_s = \left| \frac{n_1 \cos(\theta_i) - n_2 \cos(\theta_t)}{n_1 \cos(\theta_i) + n_2 \cos(\theta_t)} \right|^2, T_s = 1 - R_s \\ R_p = \left| \frac{n_1 \cos(\theta_t) - n_2 \cos(\theta_i)}{n_1 \cos(\theta_t) + n_2 \cos(\theta_i)} \right|^2, T_p = 1 - R_p \end{cases},$$

where

$$R = \frac{E_i}{E_r}, T = \frac{E_t}{E_i}$$

For **unpolarized** light,

$$R = \frac{R_s + R_p}{2}$$

It should be noted that in the case of metals, $\text{ior}(n)$ can be complex, hence the magnitude operator $\|$ in the equations.

Microgeometry

The above physics model is already enough for simulating light transport on a micro level or with perfectly smooth macro surfaces. But it fails to model situations where the surface is not perfectly smooth.

By perfectly smooth macro surfaces, we mean the surface normal has **uniform continuity**:

$$\forall \varepsilon > 0, \exists \delta > 0, (\forall \vec{x}_0, \vec{x} \in \Omega \wedge |\vec{x}_0 - \vec{x}| < \delta, |\vec{n}_0 - \vec{n}| < \varepsilon),$$

where $\Omega : F(\vec{x}) = 0$ represent the surface

In such cases, reflection & refraction can be **percieved** as having a **distribution** of directions on the macro level.

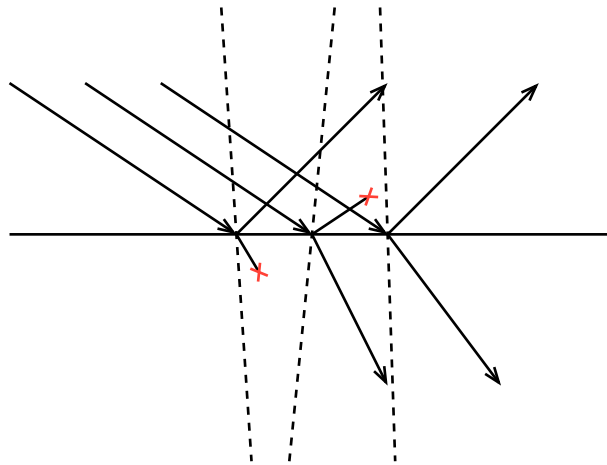


Figure 2: Rough Surface Microgeometry

The light transport behavior can be affected by the following phenomenons:

Normal Offsets

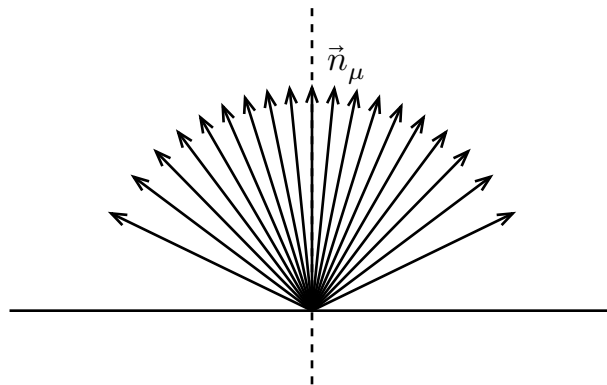
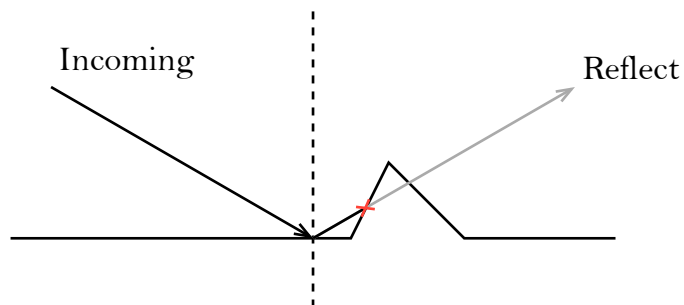


Figure 3: Normal Offsets

Masking



1.1.2 Medium Level

This section talks about what happens when light travels through some medium. For now we assume the properties of the medium doesn't change within a measurable volumn.

In the simplest scenario, when light travels through a vacume, nothing really happens. However, the complexity scales substantially when scattering & absorption happens.

1.2 Spectrum

The “percieved brightness”, a.k.a. luminance, of a spectrum can be expressed with the following integral:

$$Y = \int_{\lambda} L(\lambda) V(\lambda) d\lambda,$$

where function V denotes the spectral response curve.

For humans, which have 3 types of photoreceptor cells, there would be 3 response curves (except for the visually impaired).

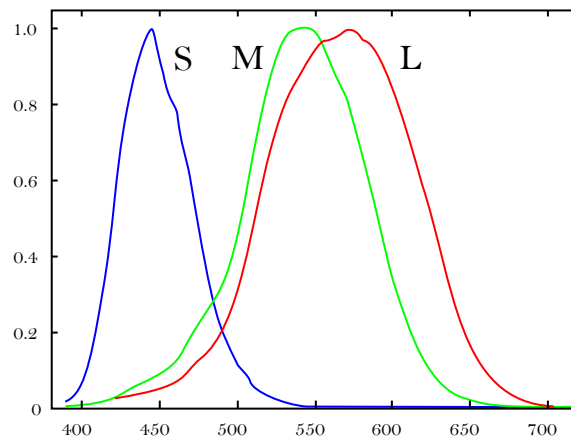


Figure 5: Human Spectral Response Curve

2 Before Implementation

This section talks about mathematical and physics tricks used in the implementation. This section **should not** contain any implementation related

information but rather some techniques/approximations that makes computing possible.

The Rendering Equation

$$L(\vec{w}_o) = L_e(\vec{w}_o) + \int_{\vec{w}_i} L(\vec{w}_i) f(\vec{w}_o, \vec{w}_i) \cos\langle\vec{w}_i, \vec{n}\rangle d\vec{w}_i$$

2.1 Light Transport Model Behavior

Generally speaking, the outward radiance $L(\vec{w}_o)$ is a cumulative product of 2 **types** of inward radiance $L(\vec{w}_i)$:

- Continuous: Diffuse
- Discrete: Specular & Transmission

Although both types are well expressed in math with the same form (Rendering Equation), we deliberately treat them differently since this form of arithmetic expression is not easily applicable in actual computation.

Continuous Models

This type of reflection(scatter) model redirects light from a **range** of inward directions to the outward direction. As the redirection involves a continuous range, a reflection(scatter) distribution function is commonly used to approximate how much a inward direction contributes to the outward direction.

$$L_o(\vec{w}_o) = \int_{\vec{w}_i} L_i(\vec{w}_i) f(\vec{w}_o, \vec{w}_i) \cos\langle\vec{w}_i, \vec{n}\rangle d\vec{w}_i$$

Discrete Models

This type of reflection(scatter) model redirects light from certain inward angles. Since the inward angles are discrete (and also of a limited number in most cases), we can just add the elements together without the need to estimate integrals 🤖.

$$L_o(\vec{w}_o) = \sum_{\vec{w}_i} L_i(\vec{w}_i) f(\vec{w}_o, \vec{w}_i) \cos\langle\vec{w}_i, \vec{n}\rangle$$

2.2 Light Transport Models

In the real world most materials exhibit combinations of reflection & scatter behaviors, but it is easier to discuss them separately since the underlying physical process is different and we use different approximation models.

2.2.1 Diffuse Reflection

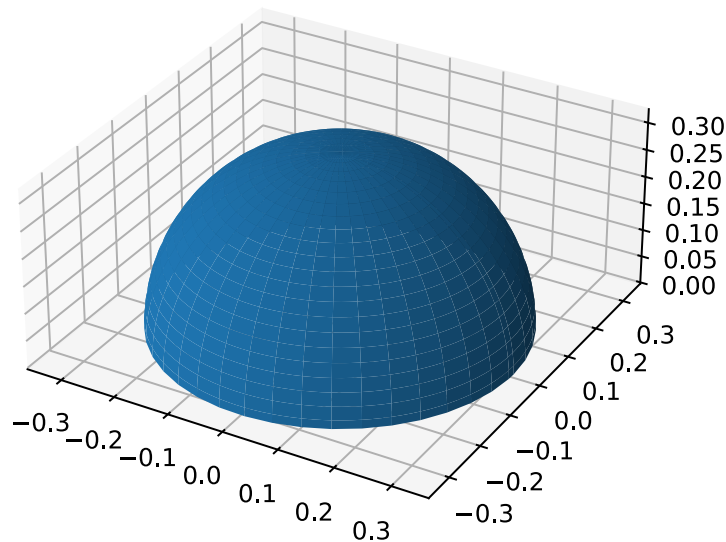


Figure 6: diffuse reflection

Lambertian Reflection

$$f = \frac{R}{\pi}$$

f is constant. And light is reflected from all inward directions in the same hemisphere.

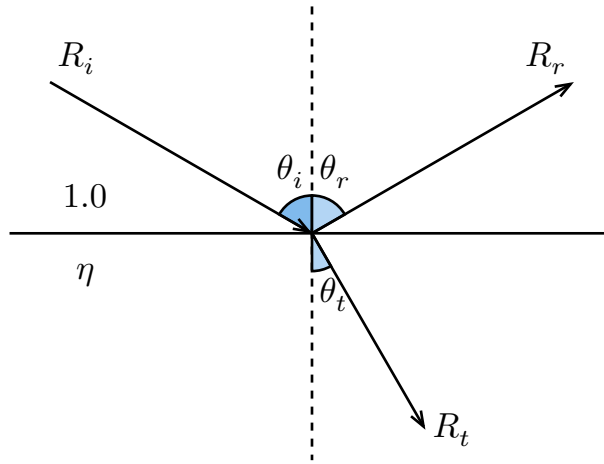
$$\text{Given } R = \iint_{\vec{w}_o} f(\vec{w}_i, \vec{w}_o) \cos \langle \vec{w}_i, \vec{w}_o \rangle d\vec{w}_o,$$

$$R = f \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \cos \varphi \sin \varphi d\theta d\varphi \quad (f \equiv C)$$

$$R = f \cdot 2\pi \cdot \frac{1}{4}$$

$$f = \frac{R}{\pi}$$

2.2.2 Specular Reflection & Refraction



We can derive these functions from the fresnel-equations:

$$\begin{cases} F_r(\omega_i) = \frac{R_r}{R_i} = \frac{1}{2} \left(\left(\frac{\eta \cos \theta_i - \cos \theta_t}{\eta \cos \theta_i + \cos \theta_t} \right)^2 + \left(\frac{\cos \theta_i - \eta \cos \theta_t}{\cos \theta_i + \eta \cos \theta_t} \right)^2 \right) \\ F_t(\omega_i) = 1 - F_r \end{cases}$$

2.2.3 Rough Reflection & Refraction

The rough reflection model approximates the behavior of micro surface changes, a.k.a. microgeometry. The model is usually represented in the form of a **normal distribution function** & a **masking function**:

$$\begin{cases} D(\vec{n}_\mu) : \int_{\Omega} D(\vec{n}_\mu) d\vec{n}_\mu = p[\vec{n}_\mu \in \Omega] (\Omega \subseteq H^2) \\ G(\omega_1, \omega_2) : \begin{cases} G(\omega_1, \omega_2) = g(\omega_1)g(\omega_2) \\ \int_{H^2} D(\vec{n}_\mu) g(\omega) \max\{0, \omega \cdot \vec{n}_\mu\} d\vec{n}_\mu = \cos \theta \end{cases} \end{cases}$$

where $\vec{n}_\mu$ denotes the micro surface normal.

It should be noted that $p(\vec{n}_\mu \in \Omega)$ means the probability of some position p on the **micro surface** having a normal in the range of Ω .

Conceptually, the masking function G bridges the gap between the micro surface area and the macro surface area, owing reductions in the micro surface area to micro level blocking.

Therefore, such distributions must have the following properties:

$$\int D(\vec{n}_\mu) (\vec{n} \cdot \vec{n}_\mu) d\vec{n}_\mu = 1$$

proof:

$$\begin{aligned} |dA| &= \int_{dA} \cos \langle \vec{n}, \vec{n}_\mu \rangle dA_\mu \\ &= \int_{H^2} \cos \langle \vec{n}, \vec{n}_\mu \rangle |dA(\vec{n}_\mu)| d\vec{n}_\mu \\ &= \int_{H^2} \cos \langle \vec{n}, \vec{n}_\mu \rangle D(\vec{n}_\mu) |dA| d\vec{n}_\mu \\ &= |dA| \int_{H^2} D(\vec{n}_\mu) (\vec{n} \cdot \vec{n}_\mu) d\vec{n}_\mu \\ &\Rightarrow \int D(\vec{n}_\mu) (\vec{n} \cdot \vec{n}_\mu) d\vec{n}_\mu = 1 \end{aligned}$$

BRDF

We then derive the complete reflection distribution function:

$$\begin{aligned} f_r(\omega_o, \omega_i) &= \frac{D(\vec{n}_\mu) G(\omega_o, \omega_i) F_r(\omega_o, \omega_i)}{4 \cos \theta_i \cos \theta_o} \\ f_t(\omega_o, \omega_i) &= \left| \frac{(\omega_i \cdot \vec{n}_\mu)(\omega_o \cdot \vec{n}_\mu)}{\cos \theta_i \cos \theta_o} \right| \frac{D(\vec{n}_\mu) F_t(\omega_i) G(\omega_o, \omega_i)}{\left(\omega_o \cdot \vec{n}_\mu + \frac{\omega_i \cdot \vec{n}_\mu}{\eta} \right)^2} \end{aligned}$$

PDF

We can also combine the 2 equations to 1 **visible normal distribution** function. This function is used to sample normals given the input angle so that the reflected ray is guaranteed to be visible:

$$D(\vec{n}_\mu \mid \omega) = \frac{D(\vec{n}_\mu)g(\omega) \max\{0, \vec{n}_\mu \cdot \omega\}}{\cos \theta}$$

Then the reflected & refracted PDF:

$$p_r(w_o) = D(\vec{n}_\mu \mid \omega_i) \frac{d\vec{n}_\mu}{d\omega_o} = \frac{D(\vec{n}_\mu \mid \omega_i)}{4(\omega_i \cdot \vec{n}_\mu)}$$

$$p_t(w_o) = D(\vec{n}_\mu \mid \omega_i) \frac{d\vec{n}_\mu}{d\omega_o} = \frac{D(\vec{n}_\mu \mid \omega_i) |\omega_i \cdot \vec{n}_\mu|}{\left(\omega_o \cdot \vec{n}_\mu + \frac{\omega_i \cdot \vec{n}_\mu}{\eta}\right)^2}$$

2.2.3.1 GGX Microfacet Model

The GGX (Ground Glass Unknown) model uses this equation as the **normal distribution function** (Trowbridge-Reitz):

$$D(\vec{n}_\mu) = \frac{1}{\pi \alpha_x \alpha_y \cos^4 \theta_\mu \left(1 + \tan^2 \theta_\mu \left(\frac{\cos^2 \varphi_\mu}{\alpha_x^2} + \frac{\sin^2 \varphi_\mu}{\alpha_y^2}\right)\right)^2}$$

Parameters

- $\alpha_x \in [0, 1]$: roughness along the x axis
- $\alpha_y \in [0, 1]$: roughness along the y axis
- θ_μ, φ_μ : inward angle

As well as a **masking function** on the assumption that $G(\omega, \omega_\mu) \equiv G(\omega)$ (Smith's approximation):

$$\int_{H^2} D(\vec{n}_\mu)g(\omega, \vec{n}_\mu) \max\{0, \omega \cdot \vec{n}_\mu\} d\vec{n}_\mu = \omega \cdot \vec{n} = \cos \theta$$

$$\Rightarrow g(\omega) = \frac{\cos \theta}{\int_{H^2} D(\vec{n}_\mu) \max\{0, \omega \cdot \vec{n}_\mu\} d\vec{n}_\mu} = \frac{1}{1 + \Lambda(\omega)}$$

where function Λ can be solved to be:

$$\Lambda(\omega) = \frac{\sqrt{1 + \alpha \tan^2 \theta} - 1}{2}, \left(\alpha = \sqrt{\alpha_x^2 \cos^2 \varphi + \alpha_y^2 \sin^2 \varphi} \right)$$

We use the approximation $G(\omega_1, \omega_2) = \frac{1}{1 + \Lambda(\omega_1) + \Lambda(\omega_2)}$ rather than $G(\omega_1, \omega_2) = g(\omega_1)g(\omega_2)$ since the latter one underestimates and the prior one appears to be more accurate in practice.

2.2.4 Light Sampling

Since rays coming from light sources contribute greatly to a surface's received radiance despite the sparse solid angles they take up, it is of great benefit to deliberately sample rays that point to the light sources rather than relying on rays sampled from the transport model probability distributions.

In this project we implemented power light sampling, which samples rays that point to light sources based on the light's strength:

Power Light Sampling

$$p(\vec{w}_i) = \frac{P_i}{\sum P},$$

where P denotes the power of light i .

Light sampling techniques only have a PDF since the BSDF used should come from the transport models.

2.3 Multiple Importance Sampling (MIS)

2.4 Spectrum

We use the CIE XYZ color space as the source of truth for outputting RGB colors.

To be more specific, the conversion process from radiance from discretely sampled wavelengths to RGB look something like this:

$$\text{RGB (Material)} \rightarrow \lambda \xrightarrow[\text{Estimator}]{\text{Monte Carlo}} \text{XYZ} \rightarrow \text{RGB}$$

2.5 Summary

Radiance

To sum up the above modeling, the complete rendering equation we simulate looks something like this:

$$\begin{aligned}
 L(\vec{w}_{\text{curr}}, p) &= L_e(p) + \int_{S^2} L(\vec{w}_{\text{next}}, p) f(\vec{w}_{\text{curr}}, \vec{w}_{\text{next}}) \cos\langle \vec{w}_{\text{next}}, \vec{n} \rangle d\vec{w}_{\text{next}} \\
 &= L_e(p) + \sum_{\text{MIS}} \int_{S^2} L(\vec{w}_{\text{next}}, p) f(\vec{w}_{\text{curr}}, \vec{w}_{\text{next}}) \cos\langle \vec{w}_{\text{next}}, \vec{n} \rangle d\vec{w}_{\text{next}} \\
 &\approx L_e(p) + \sum_{\mathcal{J}} \frac{1}{n_{\mathcal{J}}} \sum_{\vec{w}_{\text{next}}} \alpha_{\mathcal{J}}(\vec{w}_{\text{next}}) \frac{\sum_{\mathcal{T}} f_{\mathcal{T}}(\vec{w}_{\text{next}})}{p_{\mathcal{J}}(\vec{w}_{\text{curr}}, \vec{w}_{\text{next}})},
 \end{aligned}$$

where $\mathcal{J} \in \{\text{Sample Model}\} \wedge \mathcal{T} \in \{\text{Transport Model}\}$

Spectrum

And the spectrum part:

$$X = \int_{\lambda} \bar{x}(\lambda) L(\lambda) d\lambda$$

$$Y = \int_{\lambda} \bar{y}(\lambda) L(\lambda) d\lambda$$

$$Z = \int_{\lambda} \bar{z}(\lambda) L(\lambda) d\lambda$$

where $\bar{x}, \bar{y}, \bar{z}$ denote the color matching functions:

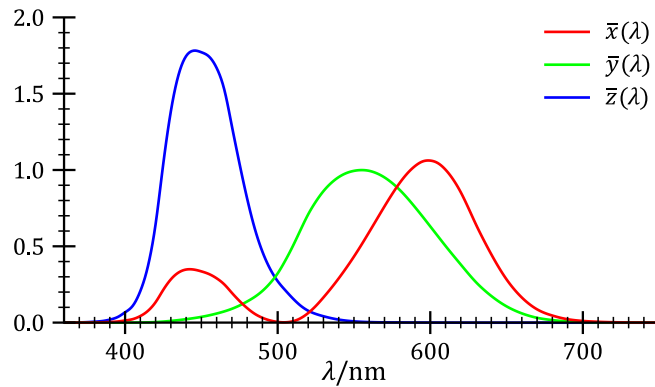


Figure 7: CIE 1931 XYZ Color Matching Functions

Then calculate the RGB values from the XYZ color space:

$$\begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{pmatrix} 2.36461385 & -0.89654057 & -0.46807328 \\ -0.51516621 & 1.4264081 & 0.0887581 \\ 0.0052037 & -0.01440816 & 1.00920446 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

source: https://en.wikipedia.org/wiki/CIE_1931_color_space

3 Implementation

3.1 Overview

The pathtracing process is divided into the following steps:

```

0  void main:
1      rays = camera.GenerateRays()
2      for ray in rays:
3          Render(ray)
4
5  Luminance Render(ray):
6      return Luminance(Trace(ray))
7
8  Radiance Trace(ray):
9      # 1. Hit Detection
10     hit_pos = ray.Hit(scene)
11     if not hit_pos.exist:
12         return 0
13
14     # 2. Sample Ray Generation
15     generated_rays = generate(ray, hit_pos)
16
17     # 3. Sample Tracing (Recursion)
18     for generated_ray in generated_rays:
19         radiance_in[ray] = Trace(generated_ray)
20
21     # 4. Integral Estimation
22     radiance_out = scatter(radiance_in)
23
24     return travel(radiance_out)

```

3.2 Hit Detection

3.2.1 Bounding Volumn Hierarchy (BVH)

3.2.1.1 Top Level Acceleration Structure (TLAS)

3.2.1.2 Bottom Level Acceleration Structure (BLAS)

3.3 Sample Ray Generation

The generation of sampling rays is done with the knowledge of the material of the hit position, as tracing them are **very expensive**. Knowing the material, we can make much higher quality samples with certain probability distributions that reduce noise and make life easier for the Scatter function (the integral estimator).

3.4 GL Transmission Format (glTF)

The glTF format is used as the accepted file format. Inorder to perform the physics based rendering techniuques from Section 1.1, we need to use the material information provided by the file format to determine which scatter model or combination of models to use. In addition, we also need to translate the material information to physics metrics to perform precise calculations. It is helpful to think of the process into 2 steps:

glTF material → physics properties → reflection models

glTF Material Information

- | | | |
|--------------|---------------|----------------|
| • alpha mode | • anisotropy | • sheen |
| • base color | • clearcoat | • specular |
| • emission | • dispersion | • transmission |
| • metallic | • ior | • volume |
| • roughness | • iridescence | |

source: <https://www.khronos.org/glTF/pbr/>

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We have this table that shows how physics properties derive from the material information:

Physics Property (Notation)	Formula
Diffuse Reflectance (R)	Smits Reconstruction $\cdot (1 - \text{metallic}) \cdot (1 - \text{transmission})$
ior (η)	$\eta = n + ki,$ where $\begin{cases} n = \begin{cases} \text{ior} & (\text{w extension}) \\ 1.5 \end{cases} \\ k_i = \text{metallic} \cdot \sqrt{\frac{4n}{1 - \frac{i}{r+b+g}} - (n+1)^2}, \\ \left(i \in \{r, g, b\} \wedge \begin{pmatrix} r \\ g \\ b \end{pmatrix} = \text{base color} \right) \end{cases}$
Microfacet Roughness ($\alpha = \alpha_x = \alpha_y$)	roughness ²

Since metal ior varies meaningfully based on the wavelength, we are actually interested in ior values on seperate rgb channels. The ior approximation is inferred from the assumption that:

- A non-metallic (metallic = 0) material should have the normal non-complex fresnel reflectance across all rgb channels:

$$F_{0,r} = F_{0,g} = F_{0,b} = F_0 \mid_{\text{metallic}=0} = \left(\frac{\eta - 1}{\eta + 1} \right)^2$$

- A fully metallic (metallic = 1) material should have the fresnel reflectance equal to the base color:

$$\begin{aligned} \left(\begin{pmatrix} F_{0,r} \\ F_{0,g} \\ F_{0,b} \end{pmatrix} \right) \Big|_{\text{metallic}=1} &= \text{base color} \\ \Rightarrow F_{0,r} + F_{0,g} + F_{0,b} &= 1.0 \end{aligned}$$

Therefore we can use this as the approximation:

$$k_i = \sqrt{\frac{4n}{1 - \text{base color}_i} - (n+1)^2},$$

and

$$\eta = n + \text{metallic} \cdot k_i$$

Inference:

$$F_{0,i}|_{\text{metallic}=1} = \text{base color}_i = \left(\frac{\eta-1}{\eta+1}\right)^2$$

$$\text{base color}_i = \left(\frac{n+k_i i-1}{n+k_i i+1}\right)^2 = \frac{(n-1)^2 + k_i^2}{(n+1)^2 + k_i^2}$$

$$1 - \text{base color}_i = \frac{(n+1)^2 - (n-1)^2}{(n+1)^2 + k_i^2} = \frac{4n}{(n+1)^2 + k_i^2}$$

$$k_i^2 = \frac{4n}{1 - \text{base color}_i} - (n+1)^2$$

Smits Reconstruction

Suppose $\{c_1, c_2, c_3\} \in \{r, g, b\} \wedge c_1 \leq c_2 \leq c_3$,

$$\text{Smits}(\lambda) = c_1 S_{\text{white}(\lambda)} + (c_2 - c_1) S_{c_2 \times c_3}(\lambda) + (c_3 - c_2) S_{c_3}(\lambda)$$

where S_c denote the spectral curve for color c and $c_2 \times c_3$ denote the color you get by mixing c_2 and c_3 . The actual curve data is extracted from the `colour` python package.

3.5 Spectrum Data Conversion

We use an approximation of the CIE color matching functions in Figure 7 for computation:

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$$\begin{aligned}\bar{x}(\lambda) &= 0.362G_s\left(\lambda; 442.0, \frac{1}{0.0624}, \frac{1}{0.0374}\right) \\ &\quad + 1.056G_s\left(\lambda; 599.8, \frac{1}{0.0264}, \frac{1}{0.0323}\right) \\ &\quad - 0.065G_s\left(\lambda; 501.1, \frac{1}{0.0490}, \frac{1}{0.0382}\right) \\ \bar{y}(\lambda) &= 0.821G_s\left(\lambda; 568.8, \frac{1}{0.0213}, \frac{1}{0.0247}\right) \\ &\quad + 0.286G_s\left(\lambda; 530.9, \frac{1}{0.0613}, \frac{1}{0.0322}\right) \\ \bar{z}(\lambda) &= 1.217G_s\left(\lambda; 437.0, \frac{1}{0.0845}, \frac{1}{0.0278}\right) \\ &\quad + 0.681G_s\left(\lambda; 459.0, \frac{1}{0.0385}, \frac{1}{0.0725}\right)\end{aligned}$$

where

$$G_s(x; \mu, \sigma_1, \sigma_2) = \begin{cases} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma_1}\right)^2} & ; x < \mu \\ e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma_2}\right)^2} & ; x \geq \mu \end{cases}$$

source: Simple Analytic Approximations to the CIE XYZ Color Matching Functions